Characteristics of Model I

1. Decision variables for each land management unit each represent a management option for the unit over the entire planning horizon.

2. Decision variables for the formulation take the form:

   X_{ij}= # of land units of management unit *i* assigned to *j*th prescription for stand *i*.

3. If rotation lengths can be short and the planning horizon is long, the number of decision variables for a single management unit can be quite large. Some systematic way of specifying the options is needed.

4. Tables like the following are often been used to describe the decision variables. For this example the minimum rotation length is two periods. Each "H" in the table represents a final harvest or the end of one rotation. Thinning options are not considered in this example.

Period	Alternative #																			
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
1														H	H	H	H	H	H	H	H
2									H	H	H	H	H								
3						H	H	H											H	H	H
4				H	H							H	H				H	H			
5			H					H			H					H					H
6		H			H		H			H			H		H			H		H	

F422 Quantitative Methods in Forest Resource Management

Model I and Model II

Model I

Definitions of Variables:

X_{aq} = the number of acres in age class a that are assigned to management sequence q.
A_a = the number of acres in age class a.
V_{aq} = the discounted net value per acre of managing a stand in age class a under management sequence q.
Y_{aqn} = the yield per acre in period n from managing a stand in age class a under management sequence q.
H_n = the harvest level in period n.
N = the number of planning periods.

Model Formulation:

Maximize $\sum_a \sum_q V_{aq} X_{aq}$ subject to

Area Constraints for Initial Conditions

$\sum_q X_{aq} = A_a$ for all a

Harvest Level Definitions

$H_n = \sum_a \sum_q Y_{aqn} X_{aq}$ for n = 1 to N

"Even Flow" Constraints ($\pm$ 10% per period)

$H_n \geq .9\ H_{n-1}$

for n = 1 to N and H_0 is given

$H_n \leq 1.1\ H_{n-1}$

Non-negativity Constraints

$X_{aq} \geq 0$ for all a, q

Note that the harvest level definition constraints could be eliminated by substituting the expression for each harvest level into the harvest (even flow) constraints. The "even flow" constraints are only one type of harvest level constraint. Other forms are possible. Constraints dealing with other management requirements are also easily incorporated. Some of the more common ones include:

- budget limitations
- non-timber resource requirements
- ending inventory requirements

Model II

Definition of variables:

X_{ij} = the number of acres that were regenerated in period i and harvested in period j.
W_{iN} = the number of acres regenerated in period i and left as part of the ending inventory.
A_i = the number of acres present in period 1 that were regenerated in period i.
M = period in which the oldest stand was regenerated.
Z = minimum harvest age.
V_{ij} = discounted net value per acre of regenerating a stand in period i and then harvesting it in period j.
Y_{ij} = yield from harvesting a stand that was regenerated in period i and harvested in period j.
E_{iN} = discounted net value per acre of stand regenerated in period i and left as standing inventory at the end of the planning period

Model Formulation:

$$\text{Maximize} \quad \sum_{j=1}^{N} \sum_{i=M}^{j-Z} V_{ij} X_{ij} + \sum_{i=M}^{N} E_{iN} W_{iN}$$

subject to

Area Constraints for Initial Conditions

$$\sum_{j=1}^{N} X_{ij} + W_{iN} = A_i \qquad \text{for all i}$$

Bookkeeping Constraints (regeneration = harvest)

$$\sum_{i=M}^{j-Z} X_{ij} = \sum_{k=j+Z}^{N} X_{jk} + W_{jN} \quad \text{for all j=1 to N}$$

Note that this set of constraints assumes that all stands are regenerated immediately as fully stocked stands.

Harvest Level Definitions

$$H_n = \sum_{i=M}^{j-Z} Y_{ij} X_{ij} \quad \text{for j=1 to N}$$

- harvest constraints—limits, even-flow, etc.—would be the same as those defined in Model I.

Non-negativity Constraints

$X_{ij} \geq 0$
$W_i \geq 0$

More about Model II

1. Decision variables for each land management unit each represent a management option for the unit over a single rotation.
2. Decision variables for the formulation take the form:

   X_{ij}= # of land units of regenerated in period *i* and final harvested in period *j*.

   Note that:
   a. the first subscript will be negative for stands in the initial inventory.
   b. the difference between the subscripts can be used to determine the age at the final harvest.
   c. more subscripts or additional variables are usually needed. For example, if all land units are not identical after harvest, one might recognize two soil types (or site index types) and use the variable sets X_{ij} and Y_{ij} or add a soil type subscript (*s*) to make the decision variable X_{sij}.
3. Model II formulations are similar to dynamic programming in that they break the decision variables into parts. A corresponding DP network could be drawn to represent the X_{ij} variables. This network will often have multiple starting nodes (defined by *i* subscript).
4. Model II formulations aggregate acres that become identical after final harvesting.
5. Model II formulations will never have fewer constraints than a Model I formulation. For problems with longer planning horizons, Model II formulations will likely have substantially fewer variables, especially if planning horizons are long.
6. Model II formulations need additional constraints to conserve acreages across rotations. In terms of the DP network, one constraint is needed for each period of final harvest. In effect, each constraint simply states that the area flowing into the bare land node must equal the area flowing out of the same bare land node.
7. As forestry problems are expanded to recognize additional facets, a key aspect is whether each facet requires additional DP networks or existing networks can just be expanded. For example, recognizing site index levels or location classes requires additional networks because sites are not similar after harvest. Differences in stocking levels can likely be incorporated into existing networks if after final harvest stocking impacts do not impact natural regeneration. Site conversions are potential linkages between networks.
8. Some claim model I formulations are easier to interpret because decision variables track each land unit for the entire planning horizon. Model II formulations offer some advantage in interpretation as they often help point out multiple optimal solutions. Model II may help make interpretations easier if there are fewer decision variables.
9. For detailed formulations where stand location relative to neighboring stands is critical, then the only advantage of a model II formulation would be in its ability to more efficiently represent a large number of management options for each individual stand. One must certainly question how many stand level options are necessary to represent the problem.